Minimizing foliations: counterexamples and common-volume rigidity

Technical summary Robert Jenkinson Alvarez

Core idea. Minimizing graph foliations at every slope do not by themselves force a periodic potential to be independent of the vertical variable. The paper constructs counterexamples in every base dimension $n \geq 2$, then identifies an exact compatibility condition: under *one common positive invariant volume*, a slope set forces $V_u = 0$ precisely when it lies on no Euclidean sphere with rational centre. The same phase calculation reconstructs every compatible family, rather than only excluding nonrigid ones.

Exact common-volume classification. Theorems 6.1 and 6.6; Proposition 6.3

For smooth solution foliations at the slopes $a \in S \subset \mathbb{R}^n$, let $p_a(x,u)$ be the gradient of the leaf through (x,u). Require a *single* positive probability density $q \in C^1(\mathbb{T}^n \times \mathbb{T})$ satisfying

$$\partial_{x_i} q + \partial_u (q p_{a,i}) = 0 \quad (a \in S,\ 1 \leq i \leq n), \qquad \int_{\mathbb{T}^n \times \mathbb{T}} q \, \mathrm{d}x \, \mathrm{d}u = 1.$$

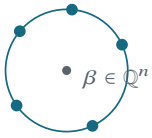
For S containing at least two distinct slopes, define

$$C_{\mathbb{Q}}(S) = \{\beta \in \mathbb{Q}^n : |a-\beta|^2 = |b-\beta|^2 \text{ for all } a, b \in S\}.$$

$$S \text{ is a universal common-volume rigidity test} \iff C_{\mathbb{Q}}(S) = \varnothing.$$

Rigidity gives $q = 1$ and $p_a = a$. Every excluded set is realized within the ordinary all-slope class. The smallest tests use **two real slopes** or **three rational slopes**; one slope never suffices. Every sufficient set contains a sufficient subset of at most $n+2$ slopes, and the bound is sharp.

A rational-centred sphere



Nontrivial compatible families exist

Three collinear rational slopes



No sphere contains them: rigidity

Two suitably chosen real slopes



$\beta_1 = 1/\sqrt{2} \notin \mathbb{Q}$: rigidity

1. The all-dimensional obstruction. The fixed Euclidean action and its equation are

$$\mathcal{A}_V[u;\Omega] = \int_\Omega (\tfrac{1}{2}|\nabla u|^2 - V(x,u))\,\mathrm{d}x, \qquad \Delta u = -V_u(x,u).$$

Ordinary deck invariance preserves the family of leaves, not necessarily each rational leaf. For a smooth periodic $b \geq 0$, nonzero but having a zero, set $V_b = -b^2/2$. If $g_E = \sqrt{b^2 + 2E}$ and $|e| = 1$, every leaf with $\nabla u = e g_E(u)$ satisfies

$$\mathcal{A}_{V_b}[w] - \mathcal{A}_{V_b}[u] = \tfrac{1}{2}\int_\Omega |\nabla w - e g_E(w)|^2 \,\mathrm{d}x.$$

This controls *all-dimensional* fixed-trace competitors. Rotating profiles give every nonzero slope; the bounded flow of $y' = b(y)$ in the x_1 direction gives slope zero.

2. The exceptional rational slope is arbitrary. The rational shear $\theta = \ell u + k \cdot x$ gives

$$V_{k,\ell}(x,u) = -\frac{\sin^4(\pi(\ell u + k\cdot x))}{2\ell^2}, \qquad \beta = -k/\ell \in \mathbb{Q}^n.$$

It has ordinary minimizing foliations at every slope, rigid foliations at every $a \neq \beta$, and *no* closed minimizing foliation at β on a finite cover. A periodic class-A minimizer of the transformed nonnegative action must be constant at a zero of its potential. Those constants cannot fill the vertical circle. Thus even one omitted rational slope can hide the holonomy obstruction.

3. A common volume locks the phases. Graph integrability and invariance make

$$d\tau_a = q(\,\mathrm{d}u - p_a \cdot \mathrm{d}x), \qquad \tau_a = u - a\cdot x + \psi_a(x,u)$$

a closed phase with periodic correction. Comparing the equations

$$\operatorname{div}_x p_a + \partial_u(\tfrac{1}{2}|p_a|^2) = -V_u$$

for two slopes, then integrating in u, makes $\psi_b - \psi_a$ harmonic on the base torus and therefore constant. Consequently

$$p_b - p_a = \frac{b-a}{q}.$$

The common density is unique for a fixed distinct pair. Separate invariant densities for different foliations are not enough; the compatibility must hold simultaneously.

4. One Fourier mode determines the obstruction. Choose a common correction ψ. Remove its vertical mean $f(x) = \int_0^1 \psi(x,u)\,\mathrm{d}u$ by setting $s = u + f(x)$ and $\chi(x,s) = \psi(x, s - f(x)) - f(x)$. Then

$$(b-a)\cdot\nabla_x\chi + \frac{|b|^2 - |a|^2}{2}\chi_s = 0, \qquad \int_0^1 \chi\,\mathrm{d}s = 0.$$

A nonzero Fourier coefficient $\widehat{\chi}(k,m)$ has $m \neq 0$ and forces, for every $b \in S$,

$$(b-a)\cdot(-k/m) = \frac{|b|^2 - |a|^2}{2}.$$

Thus $-k/m$ is one rational centre for the entire collection. If none exists, $\chi = 0$, $q = 1$ and $V_u = 0$. No differentiation or continuity in slope enters this argument.

Reconstruction, quantitative rigidity, and the nonlinear extensions

5. Every compatible family has a normal form. Let $\mathcal{L}(S)$ consist of the integer modes (k, m), $m \neq 0$, satisfying

$$(b-a)\cdot k + \tfrac{1}{2}(|b|^2 - |a|^2)m = 0 \qquad (b \in S).$$

Choose a real smooth χ with $\operatorname{supp}\widehat{\chi} \subset \mathcal{L}(S)$, zero vertical mean and $q_* = 1 + \chi_s > 0$. Put

$$v_a = \frac{a - \nabla_x \chi}{q_*}, \qquad J_a(x) = \int_0^1 v_a(x,s)\,\mathrm{d}s.$$

One mean-zero Poisson equation on the fixed base torus,

$$\Delta f = \operatorname{div}_x J_a, \qquad \int_{\mathbb{T}^n} f\,\mathrm{d}x = 0,$$

is independent of the anchor $a \in S$. With $s = u + f(x)$, reconstruct

$$\begin{aligned} \tau_a &= s + \chi(x,s) - a\cdot x, \\ q(x,u) &= q_*(x,s), \qquad p_a = v_a(x,s) - \nabla f(x), \\ Q(x,u) &= \left[\operatorname{div}_x v_a + \partial_s(\tfrac{1}{2}|v_a|^2)\right](x,s) - \Delta f(x). \end{aligned}$$

Derivatives of v_a keep s fixed. The Poisson correction gives $\int_0^1 Q\,\mathrm{d}u = 0$, so $Q = -V_u$ determines a periodic potential up to an arbitrary u-independent term. These are calibrated minimizing foliations, and *every* common-volume family arises this way. No full affine-span assumption is needed. The general reconstruction asserts foliations at S, not at all unselected slopes.

6. Full rank reduces everything to one wave. If the collection contains $n+1$ affinely independent slopes and $V_u \not\equiv 0$, their circumcentre is rational. Write it as $\beta = -k/\ell$ in primitive form. There is a nonconstant positive periodic r, normalized by $\int_0^1 r = 1$, such that

$$\begin{aligned} q(x,u) &= r(\ell u + k\cdot x), \\ V(x,u) &= A(x) - \frac{R^2}{2q(x,u)^2}, \\ p_a(x,u) &= \beta + \frac{a-\beta}{q(x,u)}, \qquad |a-\beta| = R. \end{aligned}$$

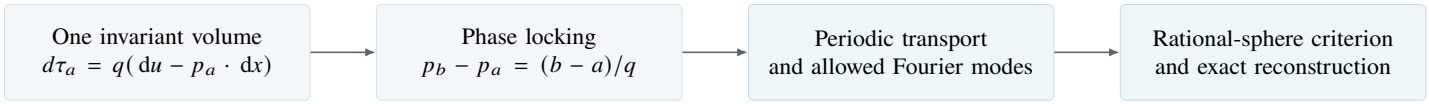
Conversely, every such r, rational β and $R > 0$ supplies the whole sphere of common-volume minimizing foliations. Below full affine rank, several independent Fourier directions may survive; a single travelling profile is no longer the complete answer.

The centre itself cannot carry a closed minimizing foliation for a nonconstant travelling potential. This gives a mixed finite test: $n+1$ affinely independent common-volume foliations, together with a closed minimizing foliation at their rational circumcentre, force $V_u = 0$. The centre foliation need not share the volume.

Sharp finite witnesses. The set

$$\{0, e_1, \dots, e_n, 2(e_1 + \cdots + e_n)\}$$

is an inclusion-minimal rational rigidity test: all $n+2$ slopes are needed, whereas every proper subset lies on a rational-centred sphere.



Rigidity: no allowed nonzero mode. Compatibility: allowed modes plus one Poisson correction.

7. A quantitative defect without common volume. For an individual admissible phase of slope a, with its own density q_a, the paper proves the exact identity

$$\begin{aligned} \mathcal{D}(\tau_a) &= \int_{\mathbb{T}^n\times\mathbb{T}} V(q_a - 1)\,\mathrm{d}x\,\mathrm{d}u \\ &= \tfrac{1}{2}\int_{\mathbb{T}^n\times\mathbb{T}} (1+q_a)|p_a - a|^2\,\mathrm{d}x\,\mathrm{d}u. \end{aligned}$$

Equality for even one phase forces $V_u = 0$. If $V_u \not\equiv 0$, a single nonzero Fourier coefficient of the force gives the uniform bound

$$\mathcal{D}(\tau_a) \geq \frac{c_V}{1+|a|^2}, \qquad c_V > 0.$$

The inverse-square order is sharp. Hence positive ensembles of phase measures can approach Haar measure for a nontrivial potential only by losing all mass from every bounded slope set. Retaining any nonvanishing bounded-slope fraction turns approximate balance into rigidity.

Rational-cell energies give a separate exact criterion: their mean-height gap vanishes precisely when a closed minimizing foliation exists. Starting from a closed foliation, a phase perturbation pins its rational slope at every nonzero amplitude. Absence of closed foliations at every rational slope is residual in the smooth topology.

8. What changes beyond quadratic kinetics. For constant anisotropic energy $K(p) = p^T G p/2$, $G > 0$, the exact criterion replaces spheres by G-ellipsoids with rational centres. For the whole class of smooth uniformly convex K, however, the universal slope criterion is different:

$$S \text{ is unbounded} \iff \text{universal rigidity at } S.$$

Every bounded S has a counterexample within that kinetic class, even with two-sided Hessian bounds and ordinary foliations at all slopes. Thus the small arithmetic tests are specific to fixed quadratic geometry, not universal over nonlinear kinetics.

For fixed real-analytic K with globally bounded positive Hessian and prescribed smooth $q > 0$, a nonlinear elliptic selector makes compatibility an *exact residual zero set*. Outside $q = 1$, $V_u = 0$, the spectrum is compact, real analytic and Lebesgue-null, with finitely many intersections with every affine line. Infinite collinear sets or positive-measure slope sets therefore force rigidity. The associated source map is proper for $q \neq 1$; in the travelling-wave sector, compatibility has a finite algebraic description through kinetic Taylor polynomials.

Scope of the higher-dimensional result. Common volume is simultaneous compatibility, not a consequence of separately existing rational foliations. Write $\mathrm{RH}_{\mathbb{Q}}$ for closed minimizing foliations at every rational slope. Independently, $\mathrm{RH}_{\mathbb{Q}}$ and $\|V_u\|_{C^{2n+6}} < \delta_n \Rightarrow V_u = 0$. The global implication from $\mathrm{RH}_{\mathbb{Q}}$ alone is not asserted. Source: §§2–7; Appendices A–H.